

Handout 2.16: Three Portfolios and a Shock

Name: _____ Date: _____ Partner: _____

Three investors each had \$1,000 in 1882 and bought at the Round 2 prices: RAIL 45, OIL 30, GAS 26, ELEC 25. Whole shares only, so a few dollars stayed in cash. Then the Panic of 1893 hit, the Round 3 prices you lived through two days ago. Then, in Part 2, they held everything to 1901, the Round 4 prices. Your job: find out what happened to each investor both times.

Risk levels for this sheet (from the company cards): ELEC high at first; RAIL medium, in a panic high; OIL medium; GAS low; BANK none.

The bank rule. The bank pays 3 percent of the balance at the start of every round, rounded to the nearest dollar. From 1882 to 1893 it paid once. From 1893 to 1901 it paid once more.

The three portfolios in 1882

Holding	Risk level	Portfolio A (the bet)	Portfolio B (the vault)	Portfolio C (the spread)
ELEC at \$25	high	20 shares = \$500	0	8 shares = \$200
RAIL at \$45	medium; in a panic, high	11 shares = \$495	0	4 shares = \$180
OIL at \$30	medium	0	0	7 shares = \$210
GAS at \$26	low	0	15 shares = \$390	8 shares = \$208
BANK	none	\$0	\$600	\$200
Cash	none	\$5	\$10	\$2
Total		\$1,000	\$1,000	\$1,000

Shock 1: the Panic of 1893 (Round 3 prices: RAIL 25, OIL 28, GAS 24, ELEC 10)

Shares times the new price equals value. The bank paid 3 percent once. Cash does not change.

Holding	New price	Portfolio A	Portfolio B	Portfolio C
ELEC	\$10	20 x 10 = ____	0	8 x 10 = ____
RAIL	\$25	11 x 25 = ____	0	4 x 25 = ____
OIL	\$28	0	0	7 x 28 = ____
GAS	\$24	0	15 x 24 = ____	8 x 24 = ____

BANK	plus 3 percent	0	$600 + 18 =$ _____	$200 + 6 =$ _____
Cash	same	5	10	2
Total after the panic		_____	_____	_____
Loss (1,000 minus total)		_____	_____	_____
Rank (1 = lost the least)		_____	_____	_____

Part 2: held to 1901 (Round 4 prices: RAIL 60, OIL 40, GAS 30, ELEC 40)

Same three portfolios, nothing sold, at the 1901 prices. The bank has now paid 3 percent twice: B's \$600 became \$618, then \$637; C's \$200 became \$206, then \$212.

Holding	New price	Portfolio A	Portfolio B	Portfolio C
ELEC	\$40	$20 \times 40 =$ _____	0	$8 \times 40 =$ _____
RAIL	\$60	$11 \times 60 =$ _____	0	$4 \times 60 =$ _____
OIL	\$40	0	0	$7 \times 40 =$ _____
GAS	\$30	0	$15 \times 30 =$ _____	$8 \times 30 =$ _____
BANK	paid twice	0	637	212
Cash	same	5	10	2
Total in 1901		_____	_____	_____
Gain (total minus 1,000)		_____	_____	_____
Rank (1 = gained the most)		_____	_____	_____

Questions

1. Which portfolio would you want in a panic? Why (use a number)?

2. Which portfolio would you want in a recovery? Why (use a number)?

3. In 1882 nobody knew the Panic of 1893 was coming. That is the real situation. Which portfolio do you pick, and what are you giving up to pick it?



4. Portfolio B barely lost and barely gained. What is one reason a real person might still want part of their money there?

My own pie (do this before Round 5)

Look at your Round 4 holdings on the ledger. Estimate the percent in each line, including the bank and cash, and draw it. Then answer.

[draw your pie here]

My biggest slice: _____ (ticker, or BANK), about _____ percent.

If my biggest slice fell by half next round, my total would drop by about \$ _____.

I will: rebalance / hold. Because:

Grade 8 stretch: Portfolio D

Design your own 1882 pie of \$1,000 in the table below (whole shares at the 1882 prices; the rest in the bank or cash) and run it through 1893 and 1901. Then compute each portfolio's percent loss and gain (loss or gain divided by 1,000, times 100).

Holding	1882 shares and amount	After the panic (1893)	Held to 1901
ELEC (25, then 10, then 40)			
RAIL (45, then 25, then 60)			
OIL (30, then 28, then 40)			
GAS (26, then 24, then 30)			
BANK (plus 3 percent each round)			
Cash			
Total	1,000		